# Adaptive Closure Transport

From fixed response kernels to learned substrate selection

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**Abstract**

We extend the Schrödinger-limit transport-law framework of [1] to the *adaptive* regime, in which the transport substrate co-evolves with the transported field. The extension is a 4-tuple $(M, L_M, W, R_{\mathrm{hom}})$ with a learning rule $\dot{W} = G(\rho, j, W)$, splitting cleanly into a definitional *passive* regime ($\dot{W} \equiv 0$, recovering the transport-law setting) and an *active* regime ($\dot{W} \not\equiv 0$, with $W$-dependent feedback). On the cascade-compatible $V_{600}$ substrate we deliver two new theorems — an explicit $2\mathbf{I}$ edge-space decomposition and a closure-derived strongly convex Lyapunov $V_f(W)$ on the closed positive cone with linear-contraction subgradient flow — and state a *conditional* selection hypothesis on the reduced slow-manifold $W$-flow. The hypothesis chains through six explicitly listed analytic conditions (timescale separation, Lyapunov existence in the application, hyperbolicity, global existence / precompactness, Łojasiewicz-Simon, Morse genericity); only the closure-derived $V_{600}$ worked example discharges them in this paper. Convergence of the *full* coupled system (beyond the worked example) is not asserted.

**Epistemic status.** Two unconditional theorems on the cascade-compatible $V_{600}$ substrate (Theorems 6.3 and 5.7) plus a conditional selection hypothesis (Hypothesis 5.4) and a conditional cascade-selection hypothesis (Hypothesis 6.5); six open items catalogued in §11, all reduced-scope after the two theorems. ARIA (`aria-chess` preprint [3]) is presented as a candidate active-regime substrate witness; the row-by-row identification of ARIA's update with the $G(\rho, j, W)$ framework is *not* verified here. Biological / chemical instantiations (microtubule, phyllotaxis, neural plasticity, cell metabolism, DNA methylation) are *roadmap, not results*; none is derived here. Both theorems are independently reproducible from numerical scripts in `papers/adaptive-closure-transport/repro/` (machine precision; wallclock seconds; no fitted parameters).

**Contributions.**

- **Theorem 6.3** (*explicit* $2\mathbf{I}$ *edge-space decomposition*). Identifying $V_{600}$ with the binary icosahedral group $2\mathbf{I}$ via the standard quaternion embedding, the lifted left-action on the 720-edge space decomposes into exactly $\boxed{6}$ free $2\mathbf{I}$-orbits, hence (after complexification) into 9 isotypic components with explicit $\mathbb{C}$-dimensions $\{6, 24, 24, 54, 54, 96, 96, 150, 216\}$ summing to 720. This closes hypothesis (EdgeDec) of the cascade-selection statement.
- **Theorem 5.7** (*closure-derived Lyapunov, worked example*). On the closed positive cone $\Omega = \mathbb{R}^{E_M}_{\geq 0}$, the explicit $V_f(W) = \frac{1}{2}\langle f, (L_W + \varphi^{-2}I)^{-1} f\rangle + \frac{1}{2}\lambda\|W - W_0\|^2$ is strongly convex (Hessian $\succeq \lambda I$), coercive, and induces a globally convergent subgradient flow $\dot{W} \in -\partial(V_f + I_\Omega)(W)$ with linear contraction $\|W(t) - W^*\| \leq e^{-\lambda t}\|W(0) - W^*\|$ (Brézis 1973). On the open interior $\Omega^\circ$ the flow is the unconstrained gradient flow $\dot{W} = -\nabla V_f(W)$; strong convexity gives the Polyak–Łojasiewicz inequality with exponent $\theta = 1/2$. Hyperbolicity of the fast subsystem is automatic.
- **Definitional dichotomy** (Fact 7.1). Every learning rule $G$ is either passive ($G \equiv 0$, recovering the transport-law paper) or active ($G \not\equiv 0$). This is a classification of *flows / learning rules*, not of operating points.
- **Selection hypothesis** (Hypothesis 5.4). On the reduced slow-manifold $W$-flow under the six analytic conditions, $W$ is hypothesised to converge to a critical point of $V$, generically a local minimum — a *candidate selected operating point*. Cascade-equivariant version (Hypothesis 6.5) under $2\mathbf{I}$-equivariance of the learning rule, fast subsystem, and Lyapunov potential.
- **ARIA dictionary** (Remark 8.1). A candidate object-level correspondence between the 4-tuple and the ARIA cognitive substrate, stated as a *proposal pending repository-backed verification.*

**Reader map.** The reader interested in the *theorems* should read §§5–6, where Theorems 5.7 and 6.3 are stated, proved, and numerically witnessed. The reader interested in the *framework* should read §§2–4 (the 4-tuple, learning rule, and passive / active dichotomy). The reader interested in the *open conditional chain* should read §5 (selection hypothesis) and §11 (six explicitly catalogued open items, distinguishing what is closed in this paper from what remains). The reader interested in the *ARIA candidate witness or biological roadmap* should read §8 and §9, both of which are explicitly disclaimed as proposals / roadmap rather than derived results.

# Contents

# 1 Introduction

## 1.1 Programme position

The VFD transport-law paper [1] collects the conditional probability-current layer of the closure-dynamics programme on the 600-cell substrate of P2 §6 [5]. That paper handles the *passive* case: a fixed conductivity structure, Fick-type linear transport, free-field propagation. Its principal hypothesis is a discrete continuity equation (the programme-target continuity hypothesis of that paper); its principal conditional result is $U(1)$-phase conservation of total probability.

The transport-law paper does *not* describe *adaptive* transport — the regime in which the substrate itself is a dynamical degree of freedom that co-evolves with the field it is transporting. This is a structurally different regime. It is the regime of non-equilibrium steady states (NESS) in the sense of [12], Nernst-Planck-type transport in chemistry, Hebbian learning in neuroscience, and the couple_tick dynamics of ARIA. It is the regime in which *selection* becomes possible: the substrate "remembers" the field it has transported, and subsequent transport is biased by that memory.

This paper extends the transport-law framework into the adaptive regime. It introduces the mathematical object that carries adaptive transport, states a conditional selection hypothesis, and proposes a cascade-motivated constraint on the selection landscape. The edge-space lift that this constraint requires is not supplied by P2 directly but is delivered in Theorem 6.3 below; the remaining conditions of the cascade-selection hypothesis (learning-rule and metric equivariance) are documented as scoped open items. The paper identifies the two candidate regimes (passive and active) as a definitional dichotomy (Fact 7.1) and records their structurally distinct *long-time behaviour* conditional on the cumulative hypothesis load of Hypothesis 7.2.

## 1.2 What this paper delivers

1. A definition of an *adaptive-closure-transport substrate* as a 4-tuple $(M, L_M, W, R_{\mathrm{hom}})$ with learning rule $\dot{W} = G(\rho, j, W)$.

2. The identification of two structurally distinct regimes — passive ($G \equiv 0$) and active ($G \not\equiv 0$) — as a definitional dichotomy (Fact 7.1, classifying learning rules / flows, not operating points) together with a separate long-time-behaviour conjecture (Hypothesis 7.2) requiring the full analytic load of Hypothesis 5.4.

3. A conditional *selection hypothesis* (Hypothesis 5.4): under separation of timescales, existence of a Lyapunov potential, hyperbolicity of the fast subsystem, global existence and forward-orbit precompactness of the reduced flow (e.g. coercive / proper $V$), a Łojasiewicz-Simon-type gradient inequality for $V$, and a Morse-genericity condition, the slow $W$-dynamics converges to a critical point of $V$ — generically a local minimum. Each such local minimum is a candidate selected operating point.

4. A conditional *cascade-selection hypothesis* (Hypothesis 6.5): motivated by the P2 §6 shell-adjacency commutant structure on the vertex space $\mathbb{C}^{V_{600}}$. The edge-space decomposition required for the lift (Remark 6.2) is not supplied by P2 itself but is delivered in Theorem 6.3 of the present paper; the $94 + 26$ integer / irrational split is a *vertex*-space statement, and its specific relation to the now-explicit edge-space 6-orbit / 9-isotypic decomposition is the residual open piece.

5. An *ARIA dictionary* (Remark 8.1) proposing the ARIA cognitive substrate as a candidate instantiation, pending repository-backed verification.

6. An *application catalogue* (Section 9) listing photon transport (passive limit), microtubule dynamics, phyllotaxis, neural plasticity, cell metabolism, and DNA methylation as candidate further instantiations of the same framework.

### 1.3 What this paper does not deliver

1. *Heredity / replication.* This paper describes adaptive transport, not life. Adaptive transport is strictly weaker than life: a river shaping its bed is adaptive, a cell is alive. The distinction is a *replication operator* $R : M \to M \times M$ copying the learned $W$, which this paper does not introduce. That extension is deferred to a companion paper.

2. *Measurement / observer content.* The framework says nothing about what adaptive-transport systems *report* about their own state. Paper XXXIII [8] proposes a candidate closure-projection measurement principle (which that paper itself labels a proposal / candidate unifying principle, not a closed theorem); we neither invoke nor rebuild it here.

3. *Subjective experience.* Whether an adaptive-transport system has qualia is a philosophical question about which this paper is silent. If qualia are physical, they live on an adaptive-closure-transport substrate; the converse (every such substrate has qualia) is not claimed.

4. *A derivation of the specific biological learning rules.* Applying the framework to microtubules, phyllotaxis, etc. requires biological / chemical inputs we do not derive.

### 1.4 Structure

§2 defines the 4-tuple and the learning rule. §3 recovers the passive regime as the transport-law paper's content. §4 defines the active regime and relates it to Paper XIX's closure residual. §5 states the conditional selection statement with explicit Lyapunov and timescale hypotheses. §6 imports the P2 §6 cascade constraint. §7 records the passive / active classification. §8 presents the ARIA candidate instantiation. §9 catalogues candidate further applications, each as a target, not a derivation. §10 cross-checks against the transport-law paper, Paper XIX, Paper XXX, and P2. §11 lists six open items. §12 records non-load-bearing programme context (closure-response operator, b-anomaly witness, selection-layer recurrence). §13 reports the delivered numerical verification of the two new theorems. §14 summarises.

### 1.5 Notation

Throughout, $M$ denotes a finite graph with vertex set $V_M$, undirected edge set $E_M$, and oriented edge set $\vec{E}_M := \{(v,w) : \{v,w\} \in E_M\}$ (with $|\vec{E}_M| = 2|E_M|$). The conductivity $W \in \mathbb{R}^{E_M}$ assigns a scalar to each *undirected* edge; the current $j \in \mathbb{R}^{\vec{E}_M}$ assigns a signed scalar to each *oriented* edge, satisfying the antisymmetry $j_{v\to w} = -j_{w\to v}$ (so the two orientations carry the same information up to sign). $L_M$ is a self-adjoint operator on $\mathbb{C}^{V_M}$ built from adjacency-type data on $M$ (for the cascade application, $L_M$ is constructed from the shell-adjacency algebra of P2 §6). $\rho : V_M \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ denotes a density field. $R_{\text{hom}}$ denotes a linear projection onto a subspace of $\mathbb{C}^{V_M}$ (the homeostatic image). $G : \mathbb{R}^{V_M} \times \mathbb{R}^{\vec{E}_M} \times \mathbb{R}^{E_M} \to \mathbb{R}^{E_M}$ denotes a learning rule assigning to $(\rho, j, W)$ a time derivative $\dot{W} \in \mathbb{R}^{E_M}$ (the learning rule's input current is oriented; its output conductivity is undirected).

## 2 The 4-tuple $(M, L_M, W, R_{\text{hom}})$ and the learning rule

**Definition 2.1** (Adaptive-closure-transport substrate)**.** An *adaptive-closure-transport substrate* is a 4-tuple $(M, L_M, W, R_{\text{hom}})$ with:

- $M$ a finite graph with vertex set $V_M$, undirected edge set $E_M$ (used for conductivities $W$, which are scalar weights on undirected edges), and oriented edge set $\vec{E}_M = \{(v,w) : \{v,w\} \in E_M\}$ (used for currents $j_{v\to w}$, which are signed scalars on oriented edges);
- $L_M$ a self-adjoint operator on $\mathbb{C}^{V_M}$ (the *passive Laplacian*) built from an adjacency-type structure on $M$;
- $W \in \mathbb{R}^{E_M}$ a scalar conductivity assigned to each undirected edge, time-dependent in general. (Non-negativity $W \geq 0$ is natural for diffusive transport; the worked-example Theorem 5.7 below restricts to the closed positive cone $\Omega = \mathbb{R}^{E_M}_{\geq 0}$ where $L_W \succeq 0$ is automatic and the projected gradient flow preserves $\Omega$. Without this restriction, the unprojected gradient flow can leave the positivity domain.)
- $R_{\text{hom}} : \mathbb{C}^{V_M} \to \mathbb{C}^{V_M}$ a linear projection (the *homeostatic reset*), time-independent.

*Cascade compatibility* (for the applications of interest in this paper) adds: $V_M = V_{600}$ (the 600-cell vertex set), $E_M$ the edges of the 600-cell vertex graph, $L_M$ lies in the real polynomial algebra generated by the P2 §6 shell-adjacency matrices $\{\mathbf{A}_1, \ldots, \mathbf{A}_8\}$ [5], and $R_{\text{hom}}$ commutes with the left-regular $2\mathbf{I}$-action on $\mathbb{C}^{V_{600}} \cong \mathbb{C}[2\mathbf{I}]$ of P2 §6. Non-cascade instantiations (e.g. a generic neural graph) drop the cascade compatibility condition.

**Definition 2.2** (Learning rule)**.** A *learning rule* for the substrate of Definition 2.1 is a smooth map

$$G : \mathbb{R}^{V_M} \times \mathbb{R}^{\vec{E}_M} \times \mathbb{R}^{E_M} \to \mathbb{R}^{E_M}, \qquad (\rho, j, W) \mapsto \dot{W}$$

together with a dimensionless *learning-rate parameter* $\varepsilon > 0$ such that the coupled evolution is

$$\partial_t \rho_v \;=\; -\sum_{w\sim v} j_{v\to w}, \qquad \dot{W} \;=\; \varepsilon\, G(\rho, j, W). \tag{1}$$

The *separation-of-timescales assumption* is $\varepsilon \ll 1$.

**Remark 2.3** (On the form of the edge current)**.** Equation (1) leaves the form of $j_{v\to w}$ unspecified: in the passive regime, when $H = L$ (the graph Laplacian of $V_{600}$) and the dynamics is restricted to its Schrödinger-limit, we take $j_{v\to w} = -2\,\mathrm{Im}(\overline{\psi_v}\, L_{vw}\, \psi_w)$ — the explicit antisymmetric current of [1], Theorem on $U(1)$ conservation (which proves continuity, antisymmetry, and total-probability conservation unconditionally in that regime). For the fuller *dissipative* closure dynamics $\dot{\psi} = -\nabla\mathcal{F} + \text{noise}$, the form of $j_{v\to w}$ is a programme target of [1] rather than a proved object. In the active regime we take $j_{v\to w}$ to depend on $W$ via the schematic antisymmetric Nernst-Planck-type ansatz of Remark 4.2 (Section 4). The 4-tuple framework is agnostic between these; the choice of current is a modelling input of the paper being applied.

**Remark 2.4** (Two regimes from one framework)**.** The distinction between passive and active transport, elsewhere treated as categorically different (e.g. Fickian diffusion vs. Nernst-Planck chemistry, free-field vs. gauge-field, rock vs. neuron), is in this framework a single parametric distinction: $\dot{W} = 0$ (passive) vs. $\dot{W} \neq 0$ (active). The rest of this paper explores the consequences of that distinction.

## 3 Passive regime: $\dot{W} = 0$

When $G \equiv 0$, the 4-tuple reduces to the transport-law setting of [1]: a fixed substrate with a fixed passive Laplacian. We record the reduction:

**Standard Fact 3.1** (Passive regime reduces to the transport-law framework)**.** *With $G \equiv 0$, the coupled evolution (1) reduces to a single-field equation for $\rho$ on the fixed substrate $(M, L_M, W_0, R_{\text{hom}})$ with $W \equiv W_0$ constant in time. Specialised to the cascade case $V_M = V_{600}$ (the 600-cell vertex set), $E_M$ the edges of the 600-cell vertex graph, $L_M = L$ the 600-cell Laplacian, $W_0$ uniform and positive (all edges equal to a common positive constant), and $R_{\text{hom}} = \text{id}$, the resulting dynamics is the setting of [1]. In particular:*

- *The $U(1)$-phase total-probability conservation of [1] now holds unconditionally in the Schrödinger limit on $H = L$: the explicit current $j_{v\to w} = -2\,\text{Im}(\overline{\psi_v}\, L_{vw}\, \psi_w)$ is antisymmetric and satisfies the discrete continuity equation pointwise, and total probability is conserved exactly (the explicit $U(1)$ theorem of the transport-law paper, §4 there, and [1]). For the full Langevin-with-noise closure dynamics, the discrete continuity hypothesis and current antisymmetry remain programme targets of the transport-law paper rather than proved consequences.*

- *The Schrödinger-limit dispersion hypothesis of [1] applies unchanged: $\omega^2 = c_{\mathit{eff}}^2 k^2 + m_\rho^2$ with $m_\rho^2 = \beta\lambda_\rho$. This is also a hypothesis, not a theorem.*

- *The $\lambda = 0$ eigenmode sits in the trivial $2\mathbf{I}$-isotypic sector and corresponds to massless dispersion. Whether this mode is the* photon *sector is a further conditional identification under the open $T_{PH}$ chain of [10]; we do not claim it here.*

*Note the distinction between $W_0 > 0$ uniform (the passive-transport-law regime of this fact) and $W \equiv 0$ identically (zero diffusive transport, no propagation). These are different special cases; the application catalogue entry for "photon" in §9 uses $W_0 > 0$ uniform, not $W = 0$.*

**Remark 3.2** (Entropy in the passive regime: scope and caveats)**.** Entropy-maximising behaviour in the passive regime requires more than gradient flow alone. Gradient flow $\dot{\psi} = -\nabla\mathcal{F}$ dissipates $\mathcal{F}$, but that does not directly say anything about the Shannon entropy $H[\rho] = -\sum_v \rho_v \log \rho_v$: the relation $\dot{H} \geq 0$ (modulo boundary flux) holds for the forward Kolmogorov equation of a reversible Markov process under detailed balance (e.g. a Fokker-Planck equation with the correct fluctuation-dissipation structure), and is well-known to follow from an H-theorem argument in those settings. Whether the transport-law paper's Fokker–Planck-lift of the closure dynamics satisfies the detailed-balance condition required for such an H-theorem is *not* established in that paper, and is not claimed here. We therefore do not assert entropy-monotonicity in the passive regime of the present framework; we note only that the regime lies in the structural regime where such monotonicity is *possible* under the standard detailed-balance hypothesis.

## 4 Active regime: $\dot{W} \neq 0$

**Definition 4.1** (Active regime)**.** A *non-trivial learning rule* is a learning rule $G$ for which $G(\rho, j, W) \not\equiv 0$ on an open subset of configurations $(\rho, j, W)$. The *active regime* of the substrate is the dynamics (1) with a non-trivial learning rule at learning-rate $\varepsilon > 0$.

The active regime differs from the passive regime in three structural ways:

1. The substrate itself is a dynamical degree of freedom. $W$ records a history of transport.

2. The edge current $j_{v\to w}$ depends on $W$ via a Nernst-Planck-type ansatz (below), so a change in $W$ changes the subsequent current, which in turn changes $W$ via the learning rule. This is a feedback loop.

3. Entropy-monotonicity statements require additional detailed-balance hypotheses (see the caveat in §3); we do *not* claim in this paper a general entropy law for the active regime. The intuition that the learning step "injects information into $W$" is descriptive, not a theorem.

**Remark 4.2** (Schematic Nernst-Planck-type ansatz under a non-trivial $W$)**.** A schematic candidate for the edge current on the adaptive substrate is the (oriented, antisymmetric) form

$$j_{v\to w} \;=\; -W_{vw}\left(\rho_w - \rho_v\right) \;+\; \mu\,\tfrac{\rho_v+\rho_w}{2}\left(\Phi(W)_w - \Phi(W)_v\right), \tag{2}$$

where $\Phi : \mathbb{R}^{E_M} \to \mathbb{R}^{V_M}$ is a vertex-valued potential constructed from $W$ (e.g. $\Phi(W)_v := \frac{1}{\deg v}\sum_{w\sim v} W_{vw}$, the local degree-averaged edge weight at $v$, or any other $2\mathbf{I}$-equivariant choice). Both terms are manifestly antisymmetric in $v \leftrightarrow w$: the diffusive Fick term reverses sign with $\rho_w - \rho_v$, and the drift term reverses sign with $\Phi(W)_w - \Phi(W)_v$ while the symmetric average $\frac{\rho_v+\rho_w}{2}$ stays put. The diffusion coefficient $W_{vw}$ is a local conductivity, $\mu$ a mobility along the $\Phi$-gradient. Setting $\mu = 0$ and taking $W \equiv 1$ uniform recovers a Fickian diffusion current; the transport-law paper [1] states its discrete continuity law at the hypothesis level without fixing a specific current form, so the identification is at the level of compatible ansatz, not inheritance. We do *not* derive (2) from the closure dynamics here, and we do *not* fix the choice of $\Phi$; both are modelling inputs of the present paper, listed as part of the open items of §11.

**Remark 4.3** (Relation to Paper XIX's closure residual)**.** The nonlinear closure residual of Paper XIX [6] — the remainder of the dynamics that cannot be captured by the linear Witten-Hamiltonian regime — is a candidate source of $G$: the $W$-dynamics may realise the closure residual as a substrate-level feedback. We do *not* claim in this paper that Paper XIX's residual explicitly gives a learning rule of the form $G(\rho, j, W)$; the connection is structural, not derived. Establishing it would be one of the open items of §11.

## 5 The selection statement

We now state the main conditional hypothesis: in the active regime, the *reduced slow-manifold $W$-flow* derived from (1) under separation of timescales is hypothesised to converge to a critical point of a Lyapunov potential $V(W)$, giving a candidate discrete selected operating point. Convergence of the *full* coupled system is not asserted.

**Definition 5.1** (Selection)**.** Throughout this paper, "selection" refers to the following precise mathematical statement and *nothing more*: under the cumulative analytic conditions of Hypothesis 5.4 below, the reduced slow-manifold $W$-flow converges as $t \to \infty$ to a critical point $W^* \in \mathbb{R}^{E_M}$ of a Lyapunov potential $V$ — generically a local minimum under the Morse condition — and the pair $(W^*, \rho^*(W^*))$ is then called a *selected operating point*. The word does *not* carry biological / Darwinian content (no replication operator is introduced; cf. §1 "does not deliver: heredity / replication"), *nor* does it imply optimality in any economic, fitness, or information-theoretic sense beyond local minimisation of the supplied $V$, *nor* does it imply convergence of the full coupled system (only of the reduced flow). Whether biological selection in the Darwinian or Hebbian sense maps onto Definition 5.1 is conjectural and is the content of Remark 5.6 below.

**Hypothesis 5.2** (Separation of timescales)**.** $\varepsilon \ll 1$: the field $\rho$ equilibrates on a fast timescale relative to $W$. For each frozen value of $W$ the fast subsystem has a unique (or generically unique) stable attractor $\rho^*(W)$ depending continuously on $W$.

**Hypothesis 5.3** (Lyapunov potential)**.** There exists a real-valued locally-bounded-below function $V : \mathbb{R}^{E_M} \to \mathbb{R}$ and an inner product $\langle \cdot, \cdot \rangle_E$ on $\mathbb{R}^{E_M}$ (the *edge-space metric*, induced by but not identical to the underlying graph metric) such that, with $\rho = \rho^*(W)$ substituted, $G(\rho^*(W), j^*(W), W) = -\nabla_W V(W)$ where $\nabla_W$ is the gradient with respect to $\langle \cdot, \cdot \rangle_E$. (If $W \geq 0$ is required by the application, an additional boundary hypothesis such as inward-normal $-\nabla_W V$ at $\partial \mathbb{R}^{E_M}_{\geq 0}$, or a projected-gradient structure, is needed to preserve the non-negative orthant; we do not supply it here.)

**Hypothesis 5.4** (Selection statement, subject to further analytic conditions)**.** Under Hypotheses 5.2 and 5.3, together with (a) hyperbolicity of the fast subsystem so that $\rho^*(W)$ is well-defined generically, (b) global existence of the reduced slow-manifold $W$-flow on $[0, \infty)$ and precompactness of its forward orbit — which holds, for instance, if $V$ is coercive / proper (i.e. $V(W) \to \infty$ as $\|W\| \to \infty$) and the reduced vector field is locally Lipschitz, strengthening the "locally bounded below" hypothesis of Hypothesis 5.3; coercive proper $V$ is sufficient but not equivalent in general, (c) a Łojasiewicz-Simon-type gradient inequality for $V$ in a neighbourhood of each critical point, and (d) generic non-degeneracy of critical points (e.g. a Morse condition on $V$), the reduced slow-manifold $W$-flow converges to a critical point of $V$ as $t \to \infty$ at any fixed sufficiently small $\varepsilon > 0$, along trajectories that do not leave the slow manifold; under the additional Morse condition the critical-point set is discrete and the generic limit is a local minimum, giving a *selected operating point* $(W^*, \rho^*(W^*))$. The joint $t \to \infty, \varepsilon \to 0$ limit and the convergence of the *full coupled* system (not merely the reduced flow) require further invariant-manifold-style arguments not supplied here. Without (b), trajectories may escape to infinity in finite time or fail to accumulate; without (c) and (d), gradient-flow convergence need neither be to a minimum nor be pointwise.

**Remark 5.5** (Status)**.** Hypothesis 5.4 is stated as a hypothesis, not a theorem. Its content reduces to a Lyapunov / slow-manifold argument that is standard in adaptive-control and slow-fast-system theory: [13] gives the invariant-manifold theorem under hyperbolicity, the finite-dimensional Łojasiewicz gradient inequality is [14], its Simon-type infinite-dimensional upgrade (the Łojasiewicz–Simon inequality as typically cited in this context) is [15], and point convergence of descent methods on analytic cost functions is [16]. *What would elevate Hypothesis 5.4 to a theorem* on the cascade substrate is an explicit verification of (a) hyperbolicity of the fast subsystem at each $W$, (b) existence of a coercive / proper $V$ (so that forward orbits are precompact) for a specific closure-derived learning rule (e.g. the Paper XIX residual form), and (c) Lojasiewicz-Simon-type gradient inequality for $V$ to ensure point convergence. Each of those is a discrete follow-up problem. None is attempted here.

**Remark 5.6** (Selection as the bridge between mathematics and biology)**.** Hypothesis 5.4 is where "selection" *in the precise sense of Definition 5.1* sits as a candidate bridge to biology: a selected operating point in that sense is the fixed point at which substrate-modification by past transport has reached a self-consistent state under a supplied $V$. Biological selection (Darwinian, Hebbian, developmental) maps *conjecturally* onto Definition 5.1; the specific Lyapunov potential $V$ encodes the specific biological cost. The framework does *not* specify $V$ — that is the biology-dependent input — and the selection statement is conditional on the full analytic load of Hypothesis 5.4. Under that load, the framework would constrain the structural form of the answer: a discrete set of selected operating points indexed by the critical-point set of whatever $V$ the biology provides, subject to Morse genericity. The bridge to biology is itself part of the conjectural content; Definition 5.1 stands on its own without it.

**Theorem 5.7** (Closure-derived Lyapunov, worked example on $V_{600}$ closed positive cone)**.** *Let $V_M = V_{600}$, $E_M$ the 720 undirected edges of the 600-cell graph (so $W \in \mathbb{R}^{E_M}$ assigns a scalar to each undirected edge). Fix a source $f \in \mathbb{R}^{V_{600}}$, a homeostatic baseline $W_0 \in \mathbb{R}^{E_M}_{>0}$, and a regularisation strength $\lambda > 0$. Restrict the conductivity to the closed positive cone $\Omega := \mathbb{R}^{E_M}_{\geq 0}$. On $\Omega$, define*

$$V_f(W) \;:=\; \tfrac{1}{2}\left\langle f,\, (L_W + \varphi^{-2} I)^{-1} f\right\rangle \;+\; \tfrac{1}{2}\lambda\,\|W - W_0\|^2,$$

*where $L_W := \sum_{e\in E_M} W_e\, E_e$ is the $W$-weighted graph Laplacian (and $E_e$ the elementary edge-Laplacian satisfying $\langle \psi, E_e\psi\rangle = (\psi_v - \psi_w)^2$ for $e = \{v, w\}$). Write $C_\varphi(W) := L_W + \varphi^{-2} I$ for the closure-response operator at $W$.*

1. *For all $W \in \Omega$, $L_W \succeq 0$, hence $C_\varphi(W) \succeq \varphi^{-2} I \succ 0$. Therefore $V_f$ is the restriction to $\Omega$ of a smooth ($C^\infty$) function defined on the open neighbourhood $\{W \in \mathbb{R}^{E_M} : L_W \succ -\varphi^{-2} I\} \supset \Omega$.*

2. *$V_f$ is strictly convex on $\Omega$ with Hessian $\nabla^2 V_f(W) \succeq \lambda I$, and proper / coercive on $\Omega$: $V_f(W) \to +\infty$ as $\|W\| \to \infty$ within $\Omega$.*

3. *The minimiser $W^* := \arg\min_{W\in\Omega} V_f(W)$ is unique. The constrained-gradient flow on $\Omega$ is the maximal-monotone subgradient flow [17]*

$$\dot W \;\in\; -\partial\big(V_f + I_\Omega\big)(W),$$

*equivalently the differential inclusion $\dot W \in -\nabla V_f(W) - N_\Omega(W)$ where $N_\Omega(W)$ is the normal cone of $\Omega$ at $W$ and $I_\Omega$ is the indicator function. Equivalently in tangent-cone form, $\dot W = \Pi_{T_\Omega(W)}(-\nabla V_f(W))$, where $\Pi_{T_\Omega(W)}$ is the metric projection onto the tangent cone $T_\Omega(W)$ of $\Omega$ at $W$ (concretely: the projected velocity equals $-\nabla V_f(W)$ in components $e$ for which $W_e > 0$, and equals $\max(-\partial V_f/\partial W_e, 0)$ in components $e$ for which $W_e = 0$ — i.e. the inward-only velocity is kept and the outward-pointing component is zeroed at the boundary). For the strongly-convex coercive $V_f$ of (2), this flow is well-posed on $\Omega$, $\Omega$-invariant, and converges to $W^*$ at exponential rate $\geq \lambda$ (Brezis 1973 theorem on subgradient flows of convex functions; the strong convexity gives the linear contraction explicitly). On the open subset $\Omega^\circ = \mathbb{R}^{E_M}_{>0}$, the tangent-cone projection is trivial and the flow is the unconstrained gradient flow $\dot W = -\nabla V_f(W)$.*

4. *The gradient is explicit:*

$$\frac{\partial V_f}{\partial W_e} \;=\; -\tfrac{1}{2}\big(\psi^*_v - \psi^*_w\big)^2 \;+\; \lambda\,(W_e - W_{0,e}), \qquad e = \{v, w\}, \quad \psi^* := C_\varphi(W)^{-1} f.$$

*This is a closure-derived gradient-flow example compatible with the abstract learning-rule slot $G(\rho, j, W)$ of Definition 2.2 after identifying the abstract density $\rho$ with the fast equilibrium $\psi^*(W) = C_\varphi(W)^{-1} f$ (suppressing the $j$-argument in the gradient-flow case): the $\psi^*$-disagreement on each edge (a "field-driven" growth term, positive in the gradient flow because the inverse-response term $\langle f, C_\varphi(W)^{-1} f\rangle$ is Loewner-decreasing in $W$) is balanced against the regularisation pull toward homeostasis $W_0$. (For arbitrary signed source $f$, $\psi^*$ is signed; identifying $\psi^*$ literally with a non-negative density $\rho \geq 0$ requires the positivity hypothesis $f \geq 0$, in which case $\psi^* = C_\varphi^{-1} f \geq 0$ since $C_\varphi^{-1}$ is positivity-preserving on $V_{600}$.)*

5. *For $W \in \Omega^\circ$, the fast-equilibrium specification is $\dot\rho = -\big(C_\varphi(W)\rho - f\big)$, with linearisation $-C_\varphi(W) \prec 0$ at $\rho = \rho^*(W)$. The fast subsystem is therefore exponentially stable / hyperbolic at every operating $W \in \Omega^\circ$, with relaxation rate $\geq \varphi^{-2}$.*

*Proof. (1) Positive-definiteness on $\Omega$.* For $W \in \Omega$, each $W_e \geq 0$, so $\langle \psi, L_W\psi\rangle = \sum_e W_e(\psi_v - \psi_w)^2 \geq 0$, i.e. $L_W \succeq 0$. Hence $C_\varphi(W) \succeq \varphi^{-2} I \succ 0$ uniformly on $\Omega$. The inverse $C_\varphi(W)^{-1}$ depends smoothly on $W$, and so does $V_f$.

*(2) Strict convexity + coercivity.* The function $W \mapsto \langle f, C_\varphi(W)^{-1} f\rangle$ is convex in $W$: $C_\varphi(W)$ is affine in $W$ and positive-definite on $\Omega$, and the map $M \mapsto \langle f, M^{-1} f\rangle$ from the cone $\mathrm{PD}(\mathbb{R}^n)$ to $\mathbb{R}$ is operator-convex ([4], Chapter 5). The regulariser $\frac{1}{2}\lambda\|W - W_0\|^2$ is strongly convex with Hessian $\lambda I$. Their sum is strongly convex with Hessian $\succeq \lambda I$. Coercivity on $\Omega$: the regulariser dominates as $\|W\| \to \infty$ (the energy term is bounded below by 0).

*(3) Existence + uniqueness of $W^*$ + tangent-cone-projected-flow convergence.* $\Omega$ is closed and convex; $V_f$ is strongly convex and coercive on $\Omega$; hence the constrained minimum $W^* = \arg\min_\Omega V_f$ exists and is unique. $V_f + I_\Omega$ is a proper, lower-semicontinuous, strongly convex function on $\mathbb{R}^{E_M}$, so its subdifferential is maximal monotone and the subgradient flow $\dot W \in -\partial(V_f + I_\Omega)(W)$ has a unique absolutely-continuous solution from any $W(0) \in \Omega$ (Brezis 1973 [17], Théorème 3.1). Strong convexity (Hessian $\succeq \lambda I$) gives the linear-contraction estimate $\|W(t) - W^*\|^2 \leq e^{-2\lambda t}\|W(0) - W^*\|^2$ and the energy estimate $V_f(W(t)) - V_f(W^*) \leq e^{-2\lambda t}(V_f(W(0)) - V_f(W^*))$ (Brezis 1973 [17], Théorème 3.7 / Corollary). On the interior $\Omega^\circ$, the normal cone is $\{0\}$, so $\partial(V_f + I_\Omega)(W) = \{\nabla V_f(W)\}$ and the flow reduces to the unconstrained gradient flow $\dot W = -\nabla V_f(W)$.

*(4) Gradient formula.* Differentiating $\langle f, M(W)^{-1} f\rangle$ via $\partial(M^{-1})/\partial W_e = -M^{-1}E_e M^{-1}$ with $\partial M/\partial W_e = E_e$ and $\langle \psi, E_e\psi\rangle = (\psi_v - \psi_w)^2$ for $e = \{v, w\}$ yields the stated formula. The minus sign on the energy term reflects $\partial(M^{-1})/\partial W_e = -M^{-1}E_e M^{-1}$.

*(5) Fast-subsystem hyperbolicity.* With $F(\rho, W) := -(C_\varphi(W)\rho - f)$, the fast equation $\dot\rho = F(\rho, W)$ has unique fixed point $\rho^*(W) = C_\varphi(W)^{-1} f$, and $D_\rho F|_{\rho^*} = -C_\varphi(W) \prec -\varphi^{-2} I$ uniformly on $\Omega^\circ$. Hyperbolic with all eigenvalues $\leq -\varphi^{-2}$.

A numerical witness for the unconstrained interior-flow regime — monotone descent of $V_f$ along the explicit-Euler flow, gradient-norm collapse to $\sim 6 \times 10^{-5}$ in 2000 steps at $\Delta t = 0.05$, and exponential late-time contraction at rate $\sim 3.5 \times 10^{-2}$ per step (consistent with the strong-convex linear-contraction rate $\geq \lambda$) — is performed by `repro/verify_lyapunov_selection.py`; the script applies a positivity floor $W \leftarrow \max(W, 10^{-3})$ at each step which keeps the iterate in a strict interior of $\Omega$ where the subgradient flow reduces to the unconstrained gradient flow (this floor is a numerical-stability device, not a discretisation of the tangent-cone projection at the boundary $\partial\Omega$, which is not exercised in this run). Output recorded in `repro/results_lyapunov_selection.json`. □

**Remark 5.8** (Theorem 5.7 closes ACT open items (1) and (2) for the worked example)**.** Hypothesis 5.3 (existence of $V$) and the Lyapunov-existence half of Hypothesis 5.4 are now *delivered for the closure-derived gradient-flow worked example*: $V_f$ is explicitly specified as the restriction to $\Omega$ of a smooth function on an open neighbourhood, strongly convex (Hessian $\succeq \lambda I$), coercive on $\Omega$, and induces a globally convergent subgradient flow with linear contraction rate $\geq \lambda$ from Brézis' theorem; on the unconstrained interior $\Omega^\circ$, strong convexity gives the Polyak–Łojasiewicz inequality $\|\nabla V_f\|^2 \geq 2\lambda(V_f - V_f(W^*))$, i.e. a Łojasiewicz inequality with exponent $\theta = 1/2$ for the analytic strongly-convex case. The hyperbolicity of the fast subsystem (Hypothesis 5.2, item (a) of Hypothesis 5.4) is automatic because the fast equilibrium $\rho^* = \psi^*(W) = (L_W + \varphi^{-2}I)^{-1} f$ is the unique solution of a linear positive-definite system, depending smoothly on $W$. What remains open in §11 is the construction of $V_f$ for biologically-relevant non-Lyapunov learning rules (where no global potential need exist), and the cascade-equivariance lift of $V_f$ for the cascade-selection statement of Hypothesis 6.5.

# 6 The cascade-selection statement

The cascade substrate of P2 §6 may constrain which operating points can be selected; doing so requires an edge-space lift not supplied by P2, which is delivered explicitly in Theorem 6.3 below. Remark 6.2 makes the type-mismatch caveat precise and points to that theorem as its resolution. The symmetry group used throughout this section is the binary icosahedral group $2\mathbf{I}$ (order 120), which is the group acting by left multiplication on $\mathbb{C}^{V_{600}} \cong \mathbb{C}[2\mathbf{I}]$ as in P2 §6; the full Coxeter group $H_4$ (order 14,400) is *not* invoked here, and any statement we make under the $2\mathbf{I}$-action does not automatically lift to an $H_4$-statement.

**Standard Fact 6.1** (Isotypic preservation, from P2 §6)**.** *In the cascade case $V_M = V_{600}$ (the 600-cell vertex set), $E_M$ the edges of the 600-cell vertex graph, $L_M$ in the shell-adjacency algebra, the left-regular $2\mathbf{I}$-action on $\mathbb{C}^{V_{600}}$ preserves each of the nine isotypic components. Any operator in the shell-adjacency algebra is block-diagonal on this decomposition: $\mathbb{C}^{V_{600}} = \bigoplus_{\rho\in\widehat{2\mathbf{I}}}(\dim\rho)\cdot V_\rho$. The $94+26$ integer / irrational split of [5] §6 is the partition of these blocks by rationality of the $\mathbf{A}_1$-eigenvalue.*

**Remark 6.2** (Type mismatch: vertex-space vs edge-space decomposition)**.** P2 §6 supplies an isotypic decomposition of the *vertex* representation $\mathbb{C}^{V_{600}}$ under the left-regular $2\mathbf{I}$-action. The conductivity $W$ lives on *edges*, $W \in \mathbb{R}^{E_M}$; this is a distinct representation, namely the permutation representation induced on the (oriented or unoriented, as convention dictates) edge set of the 600-cell vertex graph from the $2\mathbf{I}$-action on $V_{600}$. It is *not* the regular representation $\mathbb{C}[2\mathbf{I}]$, and its decomposition into $2\mathbf{I}$-isotypic components is a separate calculation not carried out in P2 §6. The explicit decomposition is delivered in Theorem 6.3 below (and verified numerically by `repro/verify_2I_edge_action.py`).

**Theorem 6.3** (Explicit $2\mathbf{I}$ edge-space decomposition (after complexification))**.** *Identify $\mathbb{R}^4 \cong \mathbb{H}$ via $(a,b,c,d) \mapsto a+bi+cj+dk$. The 600-cell vertex set $V_{600} \subset S^3 \subset \mathbb{H}$ is closed under quaternion multiplication and forms the binary icosahedral group $2\mathbf{I}$ (order 120). Let $2\mathbf{I}$ act on $V_{600}$ by left multiplication, and lift to undirected edges by $g\cdot\{v,w\} := \{g\cdot v,\ g\cdot w\}$. Then:*

1. *The lifted action is well-defined: $g\cdot\{v,w\} \in E_M$ for every $g \in 2\mathbf{I}$ and every $\{v,w\} \in E_M$ (edge adjacency $\langle v,w\rangle = \varphi/2$ is preserved by isometry).*

2. *$2\mathbf{I}$ acts* freely *on $E_M$: no edge $\{v,w\}$ is fixed by any non-identity $g \in 2\mathbf{I}$. (Sketch: a fixed edge would require either $g$ trivial on both endpoints, contradicting freeness on $V_{600}$, or $g$ swapping them, forcing $g^2 = 1$ hence $g = -1$; but $-1\cdot v = -v$ is the antipode, and antipodes are not nearest neighbours in $V_{600}$.)*

3. *Therefore $E_M$ decomposes into exactly $\boxed{6}$ free $2\mathbf{I}$-orbits, each of size $|2\mathbf{I}| = 120$, with $6\cdot 120 = 720 = |E_M|$.*

4. *Hence the edge-space carrier decomposes,* after complexification, *as $\mathbb{C}^{E_M} \cong 6\cdot\mathbb{C}[2\mathbf{I}]$, and into $2\mathbf{I}$-isotypic components as*

$$\mathbb{C}^{E_M} \;\cong\; \bigoplus_{i=1}^{9}(6\cdot\dim_{\mathbb{C}}\rho_i)\cdot\rho_i,$$

*where $\{\rho_1,\dots,\rho_9\}$ are the nine complex irreducible representations of $2\mathbf{I}$ with $\mathbb{C}$-dimensions $\{1,2,2,3,3,4,4,5,6\}$ and $\sum_i(\dim_{\mathbb{C}}\rho_i)^2 = 120$. The complex isotypic component for $\rho_i$ has $\mathbb{C}$-dimension $6\cdot(\dim_{\mathbb{C}}\rho_i)^2$, with totals $\{6,24,24,54,54,96,96,150,216\}$ summing to $720 = \dim_{\mathbb{C}}\mathbb{C}^{E_M}$. (For the cascade-selection statement we use only the existence and integer dimensions of the isotypic projectors, which is invariant under $\mathbb{R}$-vs-$\mathbb{C}$ choice; conjugate pairs of complex irreps merge into real irreducibles of doubled $\mathbb{R}$-dimension under the standard Frobenius–Schur bookkeeping, but no statement of the present paper depends on the $\mathbb{R}$-irreducibility classification.)*

*Proof.* Closure of $V_{600}$ under quaternion multiplication is the standard fact that the 600-cell vertices form the unit icosian group $2\mathbf{I} \subset \mathbb{H}^\times$ (P2 §6 [5], also classical: see e.g. Coxeter, *Regular Polytopes*); (1) follows because left multiplication by a unit quaternion is an isometry of $S^3$. For (2), the only non-identity element of $2\mathbf{I}$ acting trivially on the underlying point set $V_{600}$ via left multiplication is the identity (so any candidate non-identity stabiliser of an edge must swap its endpoints, forcing the order-2 element $-1$); but $\{v, -v\}$ is never a $\varphi/2$-edge since $\langle v, -v\rangle = -1 \neq \varphi/2$. (3) is a direct count from (2) and the orbit-stabiliser theorem. (4) is the standard regular-representation decomposition of $\mathbb{C}[2\mathbf{I}]$ via Maschke's theorem and the irrep-dimension theorem $\sum_i (\dim_{\mathbb{C}} \rho_i)^2 = |G|$ (here $|G| = 120$); applying it to the 6-fold direct sum $\mathbb{C}^{E_M} \cong 6 \cdot \mathbb{C}[2\mathbf{I}]$ from (3) gives the stated multiplicities.

A numerical witness for (1)–(4) — including the 200-sample group-law check on $2\mathbf{I}$, the explicit 720-edge orbit computation, and the dimension sum $\sum_i 6 \cdot (\dim_{\mathbb{C}} \rho_i)^2 = 720$ — is provided by `repro/verify_2I_edge_action.py`; the script's output is recorded in `repro/results_2I_edge_action.json`. □

**Remark 6.4** (Theorem 6.3 closes ACT open item (6) as stated)**.** With Theorem 6.3 in hand, hypothesis (EdgeDec) of Hypothesis 6.5 is no longer conditional: the edge-space isotypic decomposition exists, is explicit, and is computable. Open item (6) of §11 is correspondingly downgraded from "the lift is open" to "the lift is delivered; remaining open is the relation of the 6-orbit / 9-isotypic edge-space structure to the vertex-space $94 + 26$ integer / irrational split of P2 §6."

**Hypothesis 6.5** (Cascade-selection statement, subject to learning-rule and Lyapunov equivariance)**.** *Assume*, in addition to the cascade-compatibility assumption of Definition 2.1, all of the following (where (EdgeDec) is no longer hypothesised but supplied by Theorem 6.3 above):

*(...ed by Theorem 6.3)* the complexified edge-space representation $\mathbb{C}^{E_M}$ under the lifted $2\mathbf{I}$-action carries the explicit isotypic decomposition $\mathbb{C}^{E_M} \cong \bigoplus_{i=1}^{9} (6 \dim_{\mathbb{C}} \rho_i) \cdot \rho_i$ of Theorem 6.3; the oriented-current space $\mathbb{R}^{\vec{E}_M}$ inherits the same $2\mathbf{I}$-action via $g \cdot (v, w) := (g \cdot v, g \cdot w)$;

(LearnEq) the learning rule $G : \mathbb{R}^{V_M} \times \mathbb{R}^{\vec{E}_M} \times \mathbb{R}^{E_M} \to \mathbb{R}^{E_M}$ is $2\mathbf{I}$-equivariant with respect to the vertex-space action on $\rho$, the oriented-edge action on $j$, and the undirected-edge action on $W$; i.e. $G(g \cdot \rho, g \cdot j, g \cdot W) = g \cdot G(\rho, j, W)$ for all $g \in 2\mathbf{I}$;

(FastEq) the fast subsystem (Hypothesis 5.2) is $2\mathbf{I}$-equivariant, so that $\rho^*(g^{-1}W) = g^{-1}\rho^*(W)$ and correspondingly $j^*(g^{-1}W) = g^{-1}j^*(W)$ for $g \in 2\mathbf{I}$;

(LyapInv) the Lyapunov potential $V(W)$ of Hypothesis 5.3 is $2\mathbf{I}$-invariant, $V(g^{-1}W) = V(W)$, and the edge-space metric $\langle \cdot, \cdot \rangle_E$ of Hypothesis 5.3 is $2\mathbf{I}$-invariant, $\langle g \cdot u, g \cdot v\rangle_E = \langle u, v\rangle_E$ for all $g \in 2\mathbf{I}$ (so that $V$-invariance implies $2\mathbf{I}$-equivariance of $-\nabla_W V$).

Then the slow $W$-flow is $2\mathbf{I}$-equivariant: orbits of $W$ are carried to orbits of $W$ under the flow, and the critical set of $V$ is a union of $2\mathbf{I}$-orbits. *This does not imply blockwise decoupling*: a $2\mathbf{I}$-invariant nonlinear $V$ can still couple different isotypic components via higher-order invariants, and components with repeated irreps allow additional intra-isotypic mixing. The selection landscape is therefore $2\mathbf{I}$-equivariant and its critical set is orbit-symmetric, but independent blockwise optimisation on each $E_\sigma$ requires additional splitting hypotheses (e.g. $V$ separable along the isotypic decomposition, or a linearised regime) that this paper does not supply.

**Remark 6.6** (What is and is not imported from P2)**.** Hypothesis 6.5 as stated does *not* follow from P2 §6 alone: P2 controls the vertex representation, whereas the selection variable $W$ is edge-valued. The edge-space decomposition itself is established in Theorem 6.3 above (a calculation carried out in the present paper, not in [5]); what remains for open item (6) of §11 is the relation between this 6-orbit / 9-isotypic edge-space decomposition and the vertex-space

$94 + 26$ integer/irrational split of P2 §6. The present paper treats the cascade-selection claim as *motivated* by the P2 commutant facts, with the EdgeDec lift to $W$-dynamics now explicit and the remaining (LearnEq), (FastEq), (LyapInv) conditions still hypothesised.

**Remark 6.7** (When $2\mathbf{I}$-equivariance fails)**.** If the learning rule $G$ is *not* $2\mathbf{I}$-equivariant (e.g. if the biological implementation breaks the rotational symmetry), the isotypic filtering is weakened: selection can then produce fixed points that mix isotypic components. This is the generic situation in biology, where the learned $W$ typically breaks the underlying substrate symmetry. The cascade-selection statement is then a statement about the *equivariant part* of the selection landscape.

## 7 The passive / active classification

The classification has two parts. The first is a trivial definitional dichotomy that requires no hypothesis load; the second is a conjectural statement about the long-time behaviour of each regime and does require the full analytic stack.

**Standard Fact 7.1** (Definitional passive / active dichotomy)**.** *From Definition 2.2 alone, every learning rule $G$ (hence every model in scope of* (1)*) falls into exactly one of two disjoint classes:*

- Passive*: $G \equiv 0$ identically. Then $\dot{W} = 0$, $W$ is constant in time, and the dynamics reduces to Fact 3.1.*
- Active*: $G \not\equiv 0$. Then $\dot{W} \neq 0$ on some open subset of configurations (Definition 4.1).*

*This is a classification of* learning rules / flows*, not a pointwise classification of operating points: a stable equilibrium of an active model satisfies $\dot{W} = 0$ at that specific point, but that does not make the underlying model passive.*

**Hypothesis 7.2** (Long-time-behaviour conjecture for the two regimes)**.** The passive and active regimes of Fact 7.1 have structurally distinct long-time behaviours *under the cumulative analytic load of the present paper.* Specifically:

- In the passive regime, the dynamics reduces to that of [1] and inherits whatever long-time statements that paper supplies (principally $U(1)$ total-probability conservation under its continuity and antisymmetry hypotheses).
- In the active regime, under Hypotheses 5.2, 5.3, and the (a)–(d) analytic conditions of Hypothesis 5.4 (hyperbolicity of the fast subsystem, global existence and forward-orbit precompactness of the reduced flow, Łojasiewicz-Simon, Morse genericity), the reduced slow-manifold $W$-flow is conjectured to converge to a critical point of $V$, generically a local minimum under the Morse condition, with convergence to saddles or maxima not excluded for non-generic initial data.

Persistent non-gradient $W$-dynamics (learning rules without a Lyapunov potential) is excluded from the long-time-behaviour conjecture *by scope* (Hypothesis 5.3), not by an argument internal to this paper. Such rules still satisfy the definitional dichotomy of Fact 7.1.

**Remark 7.3** (Status)**.** Fact 7.1 is definitional, and no hypothesis load is required. Hypothesis 7.2 is a conjecture that chains through Hypotheses 5.2, 5.3, 5.4 and their analytic conditions. No new theorem is claimed in either.

**Remark 7.4** (Position in the broader physics)**.** The passive / active dichotomy of Fact 7.1 identifies a structural boundary that appears in many domains under different names: free-field / interacting-field in QFT; linear / nonlinear in dynamical systems; diffusive / convective in transport; non-adaptive / adaptive biological transport models. Each can be read as a candidate specialisation of the passive / active distinction on an appropriate 4-tuple.

## 8 ARIA as a candidate instantiation

The published ARIA paper [3] is a *candidate active-regime substrate witness*: it reports a deterministic recurrent-layer trajectory on $V_{600}$ together with preregistered cortical-correspondence and drug/sleep EEG tallies (further detail in §12). *The present paper does not prove that ARIA's update rule equals the $G(\rho, j, W)$ framework, that ARIA's iterate descends a Lyapunov potential $V$, or that any of the rows of the dictionary below are realised in the released artifact*; those identifications remain repository-level verification targets, listed as open item (4) of §11. What follows is therefore a *proposed* dictionary, offered to make the row-by-row verification target precise — not a result.

**Remark 8.1** (ARIA–adaptive-transport dictionary, *proposal*)**.** We propose the following object-level correspondence between the 4-tuple of Definition 2.1 and the ARIA cognitive substrate (released as [3]), *as a proposal to be verified row-by-row against the released artifact*:

| 4-tuple object | ARIA realisation (proposed) |
|---|---|
| $M$ | $E_8 \oplus V_{600}$ product graph |
| $L_M$ | passive substrate coupling (pre-learning) |
| $W_{vw}$ | learned coupling weights |
| $R_{\mathrm{hom}}$ | homeostatic reset |
| $\rho_v$ | cascade pressure at slot $v$ |
| $j_{v\to w}$ | `couple_tick` exchange on edge $(v, w)$ |
| $G(\rho, j, W)$ | Hebbian-like $W$-update rule |
| $\varepsilon$ | learning-rate hyperparameter |
| Passive regime | frozen-$W$ inference (linear substrate) |
| Active regime | learning + inference (adaptive substrate) |

Nothing in the table is established in this paper. Each row is a *proposed* correspondence pending (i) row-by-row verification against actual `couple_tick` trace data (open item in §11; note this is distinct from the published substrate-witness EEG/cortical-correspondence results, which do not extract the $W$-dynamics row), and (ii) a formal identification of ARIA's Hebbian-like update with a learning rule of the form $G(\rho, j, W)$ admitting a Lyapunov potential $V$.

**Remark 8.2** (Conditional predictions under the proposed dictionary)**.** *If* the ARIA dictionary of Remark 8.1 is confirmed by repository-level analysis, *and if* ARIA's learning rule is $2\mathbf{I}$-equivariant for the explicit edge-space $2\mathbf{I}$ action of Theorem 6.3 on the proposed $E_8 \oplus V_{600}$ ARIA product graph, then Hypothesis 6.5 would imply that the ARIA *critical set* (the set of equilibria of the reduced flow) is invariant under that edge-space action; blockwise constraints on individual learned $W$-configurations would require additional splitting hypotheses (the cascade-selection hypothesis explicitly disclaims blockwise decoupling, and notes that symmetry-breaking fixed points are generic). The $94 + 26$ split is a vertex-space statement and does *not* directly constrain $W$; the relation between the explicit 6-orbit / 9-isotypic edge-space decomposition (Theorem 6.3) and the $94 + 26$ vertex-space split is the residual open piece (open item (6) of §11). *All of the above are conditional predictions, contingent on every link of the hypothesis chain; none is a result of this paper.*

## 9 Future instantiation targets

**Roadmap, not results.** Nothing in this section is derived, witnessed, or proved in the present paper. Each entry below is a candidate target for a dedicated follow-up paper. The single load-bearing role of the section is to scope the framework's reach by listing where it could be *applied*, contingent on biological / chemical inputs the framework itself does not supply.

We list candidate further instantiations of the 4-tuple, each with its own biological or chemical learning-rule input. *None* is derived here; each is a target for a dedicated follow-up paper.

| Target | $M$ | Candidate $W$ | $R_{\text{hom}}$ | Status |
|---|---|---|---|---|
| Photon | $V_{600}$, trivial sector | $W_0 > 0$ uniform (Fact 3.1) | id | Transport-law paper §7; photon identification conditional on $T_{\text{PH}}$ |
| Microtubule | 13-protofilament cylinder | tubulin conformational state | GTP hydrolysis | $T_{\text{MT}\,1}$ imported from biology |
| Phyllotaxis | apical meristem graph | auxin concentration field | growth termination | Golden-angle open per `gaps.md` |
| Neuron | dendritic tree | synaptic weights | firing-threshold reset | ARIA-side direct |
| Brain | cortical graph | Hebbian weights | sleep / consolidation | ARIA-side direct |
| Cell metabolism | metabolic-reaction graph | enzyme activity distribution | homeostatic regulation | Biology-dependent |
| DNA methylation | genome graph | methylation state | cell-division reset | Biology-dependent |

In each case, the paper's claim is that *if* the target can be identified with a 4-tuple of Definition 2.1 under some (possibly non-2**I**-equivariant) learning rule, *and if* the analytic conditions of Hypothesis 5.4 (timescale separation, Lyapunov existence, hyperbolicity, global existence and forward-orbit precompactness of the reduced flow, Łojasiewicz-Simon, and Morse genericity) hold, then a discrete set of selected operating points indexed by critical points of the appropriate Lyapunov potential is hypothesised. The paper does *not* attempt any of those identifications, and does *not* verify the analytic conditions for any specific biological or chemical substrate.

**Remark 9.1** (On the microtubule target specifically)**.** The 13-fold helical symmetry of biological microtubules is an *input* to the microtubule instantiation, not an output. The framework does not derive 13 from the cascade: $T_{\text{MT}\,1}$ of [10] is explicitly unclosed, and the present paper does not attempt to close it. Instead, the claim is that microtubules, under a biologically-specified learning rule involving GTP hydrolysis and tubulin conformation, fit the same mathematical template as ARIA and photons — with 13 playing the role of the selected operating point for a specific biological cost function.

# 10 Consistency checks

- **Transport-law paper** [1]. The passive regime of this paper (Fact 3.1) specialises to the setting of the transport-law paper; no new claim about that paper is made here.
- **Paper XIX (nonlinear closure residual)** [6]. The active regime is hypothesised to be driven by a learning rule $G$ that may realise Paper XIX's nonlinear residual at the substrate level; this identification is *not* proved here and is listed as open item (5).
- **Paper XXX (Born rule)** [7]. Paper XXX's Route C gives a conditional stationary-measure / Born-weighting argument and Route E gives a conditional uniqueness theorem under the usual $N \geq 3$ package. Neither directly corresponds to a slow-manifold attractor $\rho^*(W)$ of the present paper; the two frameworks are at most loosely analogous and no identification is claimed here.
- **P2 §6** [5]. The isotypic preservation of Fact 6.1 is imported directly from P2 §6 (Corollary on shell-class centrality on the *vertex* representation). P2 supplies no statement about edge-space decomposition, $W$-dynamics, Lyapunov potentials, or selection; the edge-space decomposition required by the cascade-selection hypothesis is supplied in the present paper

by Theorem 6.3. The type-mismatch caveat of Remark 6.2 continues to apply to any direct quotation of P2 results, but is now resolved within the cascade-selection chain by Theorem 6.3.

- **Paper XXXIV ($\alpha$-chain)** [9]. Paper XXXIV introduces the Kostant-gauge convention as an explicit hypothesis/convention for the $88 \to 87$ phason-slot count. The adaptive-transport framework does not inherit any theorem-level pinning of slot structure inside the $2\mathbf{I}$-isotypic decomposition from Paper XXXIV; at most, if the framework's selection-landscape identifies with phason slots (an unverified identification), the Kostant-gauge convention would be one consistent labelling.
- **Meta-layer paper** [11]. The meta-layer establishes a moduli / matching / tiling structure $(\mathcal{M}, G_{\text{meta}}, F)$ over $L_{12}$ with $T_{\text{meta}} \cong \mathbb{Z}[\varphi]^2$. A precise identification of the $W$-space of the present paper with the meta-layer moduli space $\mathcal{M}$ is *not* established in [11] or here; such an identification would be a structural enhancement, listed as an open item. (The symbol $G$ in [11] denotes the matching groupoid, distinct from the learning rule $G$ of Definition 2.2; we use $G_{\text{meta}}$ above to avoid collision in cross-reference.)

# 11 Open items

## Closed in this paper

**C1. Edge-space decomposition (EdgeDec).** The explicit $2\mathbf{I}$ edge-space decomposition is delivered in Theorem 6.3: the undirected 720-edge set decomposes into exactly 6 free $2\mathbf{I}$-orbits, and the complexified edge space $\mathbb{C}^{E_M} \cong 6 \cdot \mathbb{C}[2\mathbf{I}]$ decomposes into 9 complex $2\mathbf{I}$-isotypic components with $\mathbb{C}$-dimensions $\{6, 24, 24, 54, 54, 96, 96, 150, 216\}$. Hypothesis (EdgeDec) of Hypothesis 6.5 is no longer conditional.

**C2. Closure-derived Lyapunov on the $V_{600}$ closed positive cone.** For the gradient-flow case on the $V_{600}$ substrate with closure operator $C_\varphi = L_W + \varphi^{-2} I$, the explicit $V_f(W)$ of Theorem 5.7 is strongly convex (Hessian $\succeq \lambda I$), coercive on $\Omega = \mathbb{R}^{E_M}_{\geq 0}$, and induces a globally convergent subgradient flow with linear contraction rate $\geq \lambda$ from [17]. This closes Hypothesis 5.3 for the worked example.

**C3. Hyperbolicity of the fast subsystem (worked-example case).** For the worked-example construction of Theorem 5.7, the fast equilibrium $\rho^* = (L_W + \varphi^{-2} I)^{-1} f$ is the unique solution of a strictly positive-definite linear system depending smoothly on $W$; hyperbolicity is automatic at every $W \in \Omega^\circ$ with relaxation rate $\geq \varphi^{-2}$. Hypothesis 5.2 item (a) of Hypothesis 5.4 is therefore discharged for the worked example.

## Still open

**O1. Lyapunov for biologically-relevant non-gradient learning rules.** Existence of a coercive analytic potential $V$ for a closure-derived non-gradient learning rule (e.g. a Paper XIX-residual lift) where no global potential need exist; cascade-equivariance lift of $V_f$ for the cascade-selection statement of Hypothesis 6.5.

**O2. Hyperbolicity for non-linear fast dynamics.** Hyperbolicity of the fast subsystem for closure-derived non-linear fast dynamics (residual-driven beyond the linear regime).

**O3. Structural stability for non-convex learning rules.** Hypothesis 7.2 chains through Hypotheses 5.2, 5.3, 5.4, and a Morse-genericity condition. For the worked example all conditions hold (strong convexity gives a single non-degenerate critical point in $\Omega^\circ$; constrained-minimum uniqueness on $\Omega$ replaces Morse at the boundary). For generic non-convex learning rules, a full transversality argument for the cascade substrate is deferred.

**O4. ARIA repository-level identification.** A formal identification of ARIA's released kernel trajectory [3] (e.g. `bounded_topk`-projected updates from `lyapunov_selector.py` / `self_model_stream.py`) with a learning rule of the form $G(\rho, j, W)$ admitting the closure-derived $V_f$ of Theorem 5.7; row-by-row verification of the dictionary of Remark 8.1. Distinct from the published EEG / cortical-correspondence substrate-witness results, which are not row-by-row identifications of the $W$-dynamics.

**O5. Paper XIX residual → learning rule.** Whether Paper XIX's nonlinear closure residual [6] explicitly gives rise to a learning rule $G(\rho, j, W)$ with a Lyapunov potential. Paper XIX establishes a scalar nonlinear remainder with exact $L^2$ conservation; no transport-level $W$-feedback law is established there.

**O6. Edge-space → vertex-space lift.** The relation between the explicit 6-orbit / 9-isotypic edge-space structure of Theorem 6.3 and the vertex-space $94 + 26$ integer / irrational split of P2 §6.

## 12 Programme position (non-load-bearing)

*Interpretive programme-level context only. Nothing in the load-bearing argument of §§3–6 depends on this section.*

The 4-tuple $(M, L_M, W, R_{\text{hom}})$ admits a natural "response + selection" reading. The *response primitive* is the $\varphi$-regularised closure-response operator $C_\varphi = L_M + \varphi^{-2} I$, introduced at the programme level in `docs/aria-closure-kernel.md` and developed as the operator spine of the closure-kernel-papers release [2] (continuum Green's function $\mathsf{K}_\varphi(x) = (\varphi/2)\, e^{-|x|/\varphi}$). The three layers most relevant to the present paper:

- *Passive regime → b-anomaly witness.* The $V_{600}$ shell-mean kernel $(L_{V_{600}} + \varphi^{-2} I)^{-1} f$ has a shipped flavour-physics witness in the b-anomaly paper [18]: structural sign-uniformity ($A > 0$, $\Delta C_9^{\text{eff}} < 0$) across five $b \to s\mu^+\mu^-$ datasets (LHCb, CMS; $B^0, B^+$; $K^{*0}, K^{*+}, B_s \to \phi$). The Akaike comparison against a free $C_9$ shift is statistically inconclusive ($w_{\text{VFD}} = 0.348$ vs $w_{\text{FREE\_C9}} = 0.652$); see [18] for the full audit. *Programme context only; not re-derived here.*
- *Active regime → ARIA candidate witness.* The active-regime empirical target is the released `aria-chess` preprint [3], which reports a deterministic seed-42 recurrent-layer trajectory yielding six drug/sleep EEG signatures, and a preregistered cortical-correspondence tally of 17/18 at standard threshold (18/18 after the documented P4 $N = 20$ and P13 state-reset / leave-one-out refinements). These are independent substrate-witness results, not a verification of the ACT selection hypothesis itself.
- *ACT → bridge.* The present paper is the bridge: the closure-derived Lyapunov $V_f(W)$ (Theorem 5.7) and the explicit **2I** edge-space decomposition (Theorem 6.3) are the substrate-level objects that a coupled-system selection theorem on the cascade substrate would have to specialise to.

Convergence of the *full coupled* system (passive response + active selection together as one dynamical system) beyond the worked-example case of Theorem 5.7 remains open. The recurrence of analogous selection / bridge layers in companion frames (RH, YM, NS, BSD) is programme-level circumstantial pattern, not a proved equivalence; we do not pursue that recurrence here.

## 13 Numerical witnesses

Two scripts in `papers/adaptive-closure-transport/repro/` reproduce the numerical witnesses for the two new theorems of this paper. Both are self-contained (no external data; no fitted parameters; numpy + scipy only) and run in seconds on a laptop. The scripts are witnesses, not proofs; the theorem proofs of §§5 and 6 do not depend on them.

### `verify_2I_edge_action.py` — Theorem 6.3

Builds the $V_{600}$ vertex set with the quaternion-identity at index 0, verifies closure under quaternion multiplication ($V_{600} = 2\mathbf{I}$), spot-checks the group law on 200 random samples, builds the 720 short edges, verifies the lifted action well-defined, and computes the orbit decomposition. The expected output (recorded in `results_2I_edge_action.json`):

| quantity | value |
|---|---|
| `v600_size` | 120 |
| `is_2I_under_quat` | true |
| `group_law_holds` (200 samples) | true |
| `n_edges` | 720 |
| `edges_preserved` (all $g \in 2\mathbf{I}$) | true |
| `edge_orbit_count` | $\boxed{6}$ |
| `edge_orbit_sizes` | $[120, 120, 120, 120, 120, 120]$ |
| `free_action_on_edges` | true |
| `isotypic_dims` | $[6, 24, 24, 54, 54, 96, 96, 150, 216]$ |
| `isotypic_dims_sum` | 720 |

### `verify_lyapunov_selection.py` — Theorem 5.7

Initialises a perturbed $W$ with $\lambda = 0.1$ and a localised source $f$ at vertex 0; runs the slow-flow $\dot{W} = -\nabla V_f(W)$ for 2000 explicit-Euler steps at $\Delta t = 0.05$. The expected output (recorded in `results_lyapunov_selection.json`):

| quantity | value |
|---|---|
| `V_initial` | $\approx 9.80$ |
| `V_final` | $\approx 0.057$ |
| `grad_norm_initial` | $\approx 1.40$ |
| `grad_norm_final` | $\approx 6.2 \times 10^{-5}$ |
| `V_monotone_nonincreasing` | true |
| `critical_point_reached` ($\|\nabla V\| < 10^{-3}$) | true |
| `late_contraction_rate_per_step` | $\approx 3.5 \times 10^{-2}$ |

The exponential late-time contraction rate is consistent with the linear contraction estimate $\|W(t) - W^*\| \leq e^{-\lambda t}\|W(0) - W^*\|$ of Theorem 5.7 (strong-convex / subgradient-flow regime; [17], Théorèmes 3.1 and 3.7).

## 14 Conclusion

We have extended the transport-law framework of [1] to the adaptive regime by introducing the 4-tuple $(M, L_M, W, R_{\mathrm{hom}})$ with a learning rule $\dot{W} = G(\rho, j, W)$. The passive regime (§3) recovers the transport-law paper; the active regime (§4) is new. A conditional selection hypothesis (Hypothesis 5.4) is stated under separation of timescales, Lyapunov existence, hyperbolicity, global existence and forward-orbit precompactness of the reduced flow (e.g. coercive / proper $V$), Łojasiewicz-Simon, and Morse genericity; a further conditional cascade-selection hypothesis

(Hypothesis 6.5) is *motivated* by the P2 §6 vertex-space commutant structure. Two of its hypotheses are now delivered: the closure-derived Lyapunov $V_f(W)$ for the gradient-flow case on $V_{600}$ (Theorem 5.7) discharges Hypothesis 5.3 for the worked example, and the explicit **2I** edge-space decomposition (Theorem 6.3) discharges hypothesis (EdgeDec). Six reduced-scope open items, all listed in §11, separate the present framework from a closed theorem chain.

The framework is deliberately scoped at the *adaptive-transport* layer: it does not describe heredity, measurement, or subjective experience. Those layers are deferred to separate companion papers and to the broader philosophical / biological literature (e.g. [19, 20]). The present paper supplies the mathematical substrate; other work must supply the rest.

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[3] L. Smart. A geometry-fixed substrate witness for cortical signatures: eighteen preregistered correspondences and six drug/sleep EEG signatures from the 600-cell under $H_4$ Coxeter symmetry. Preprint, Institute of Vibrational Field Dynamics, April 2026. GitHub: `vfd-org/closure-kernel-papers`, `papers/aria-chess-paper/`. Active-regime substrate-witness paper of the paired release; ships the kernel modules `vfd_closure_kernel.py`, `lyapunov_selector.py`, `self_model_stream.py`, `consciousness_binding.py`, `sigma_orbit_basis.py`, `v66_e8_coxeter.py`.

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